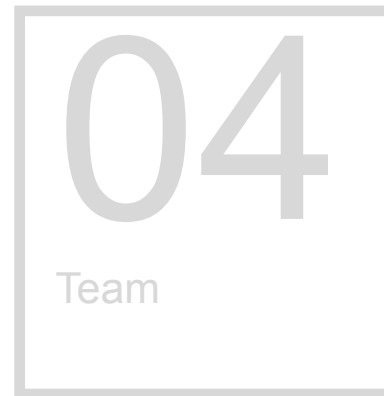
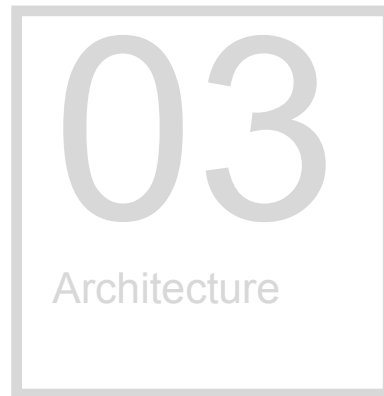


Build With AI for Sustainable Growth - Caucasus & Central Asia 2024

by AirWatch.ai



Content



Problem: air pollution

10,400 people die annually due to air pollution-related diseases in Kazakhstan ⁽¹⁾

Who is experiencing the problem?

Most polluted countries and regions by annual average PM2.5 concentration ($\mu\text{g}/\text{m}^3$) ⁽²⁾

Rank	Country/Region	2023	2022	2021	2020	2019	2018	Population
1	Bangladesh	79.9	65.8	76.9	77.1	83.3	97.1	169,356,251
2	Pakistan	73.7	70.9	66.8	59	65.8	74.3	231,402,117
3	India	54.4	53.3	58.1	51.9	58.1	72.5	1,407,563,842
40	Kazakhstan	22.2	23	31.1	21.9	23.6	29.8	19,000,988
132	Iceland	4	3.4	6.1	7.2	5.5	5	372,520

PM2.5 Legend 0-5: Complies with WHO standards | 5.1-10: Exceeds by 1-2 times | 10.1-15: Exceeds by 2-3 times | 15.1-25: Exceeds by 3-5 times | 25.1-35: Exceeds by 5-7 times | 35.1-50: Exceeds by 7-10 times | >50.1: Exceeds by more than 10 times

Diseases from air pollution:

1. Cardiovascular Diseases: heart attacks (myocardial infarction), stroke hypertension (high blood pressure).
2. Respiratory Diseases: chronic obstructive pulmonary disease (COPD), asthma, lung cancer.
3. Premature Birth and Low Birth Weight: complications in pregnancy
4. Diabetes.
5. Neurodegenerative Diseases: Alzheimer's disease and other Dementias.

Latest **World Bank** research shows that air pollution in only four regions of Kazakhstan - Almaty, Karaganda, Pavlodar and East Kazakhstan – causes ⁽³⁾:

1. 1,016 new cases of chronic bronchitis;
2. 2,848 premature deaths;
3. 2,665 deaths from cardiopulmonary disease;
4. 183 deaths from lung cancer.



Source:

1. <https://www.worldbank.org/en/news/press-release/2023/04/25/kazakhstan-discusses-actions-to-reduce-air-pollution-with-partners>

2. <https://www.igair.com/ru/world-most-polluted-countries>

3. https://www.kt.kz/rus/ecology/v_kazhastane_rastet_smertnost_iz-za_gryaznogo_vozduha_-_1377949515.html

Scale of problem: air pollution in Kazakhstan

Kazakhstan loses **2.5 billion \$** annually due to polluted air.

Who is experiencing the problem?

In general, **most of southern Kazakhstan**:

1. **Almaty**: The city experiences some of the highest levels of air pollution in Kazakhstan. In winter, the concentration of PM2.5 can reach levels that are 10 times higher than the World Health Organization (WHO) recommended limits. Population size: 2.2 mln people.

2. **Astana and Shymkent**: These cities also suffer from elevated levels of air pollution, particularly during the winter months due to coal heating and increased vehicle emissions.

Population size in two cities: 2.6 mln people.

3. **Temirtau, Balkhash, and Zhezkazgan**: These industrial towns are hotspots for air pollution due to heavy industries like steel production and mining. In Temirtau, for instance, mortality rates due to air pollution-related diseases are significantly higher than in other parts of the country, with rates ranging from 273 to 373 deaths per 100,000 adults annually.

Population size in three cities: 0.35 mln people.

The economic impact of air pollution on public health in specific cities in Kazakhstan

The damage due to the impact of polluted air on public health in some cities in 2011, excluding premature deaths caused by air pollution, amounted to \$486 million in Almaty, \$181 million in Pavlodar, \$157 million in Ust-Kamenogorsk, and \$154 million in Temirtau. The republican damage due to diseases caused by air pollution is **2.5 billion dollars annually, which is 1.7% of GDP** ⁽²⁾

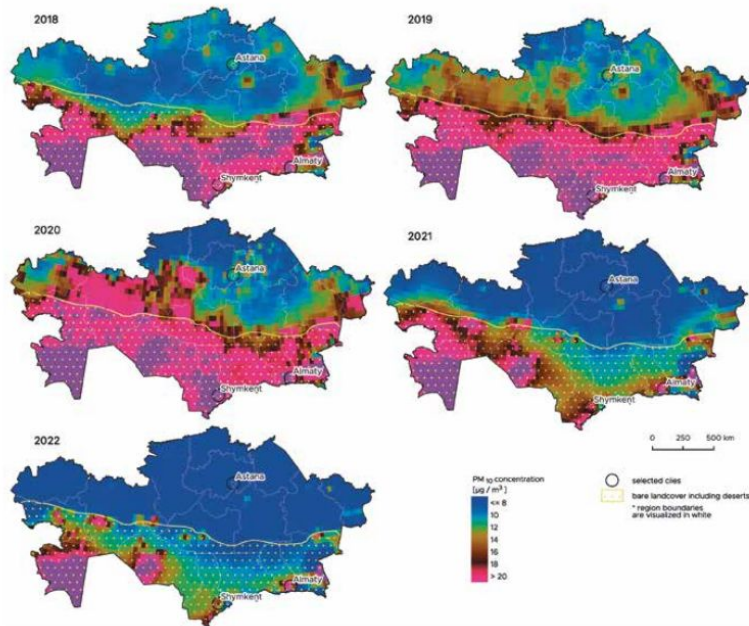
Current solutions:

1. Regulatory Measures:

- Environmental Code: Kazakhstan's Environmental Code, which was amended in 2021, provides a legal framework for air quality management.

2. Public and Private Initiatives:

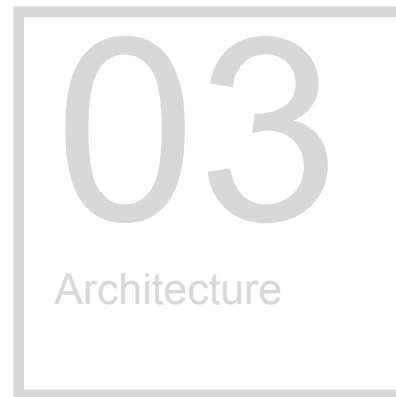
- Green Technology Projects: Both public and private sectors are investing in green technologies to reduce emissions. This includes the development of electric public transport like in Astana & Almaty.
- Awareness Campaigns: Almaty Air.



Source:

1. Picture: Copernicus Atmospheric Monitoring Service data (CAMS, 2022); OpenStreetMap contributors, 2022; European Commission, 2022
2. Aigul Sagadibekovna Solovyova

Content



Product: value proposition

AirWatch.ai is a web app with artificial intelligence-based air quality monitoring system that uses Gemini AI technology to provide real-time air quality data and predict air quality changes. The system is designed to provide timely alerts to pedestrians, residents of homes along roadways, and businesses with access to roadways about necessary measures to reduce exposure to air pollution. This project contributes to improving the health and safety of citizens in the city.

UN Sustainable Development Goals (SDGs) and Targets:

1

1. SDG 3: Good Health and Well-being: This goal aims to ensure healthy lives and promote well-being for all ages. AirWatch.ai contributes directly to this by providing real-time air quality data and predictive analytics, which help reduce exposure to harmful pollutants.

2

2. SDG 11: Sustainable Cities and Communities: This goal focuses on making cities inclusive, safe, resilient, and sustainable. AirWatch.ai contributes to this goal by enabling city residents, particularly those living near roadways, to take timely measures to protect themselves from air pollution.

3

3. SDG 13: Climate Action: This goal emphasizes the need to combat climate change and its impacts. By providing data on air quality and predicting pollution trends, AirWatch.ai helps mitigate the effects of pollution, a key factor in climate change.

Short-Term Goals (During the Hackathon):



Prototype Development:

- Objective: Develop a fully functional prototype of the AirWatch.ai web app.
- Metrics: Complete the integration of Gemini technology for real-time air quality monitoring and predictive analytics.
- Outcome: Demonstrate the app's capability to provide real-time air quality data and issue alerts to users based on current pollution levels in a selected city (e.g., Almaty).

Initial User Engagement:

- Objective: Conduct a small-scale pilot test with a target group (e.g., 100-200 users in a specific area).
- Metrics: Track the number of users who sign up and actively use the app during the hackathon.
- Outcome: Gather initial user feedback to identify potential improvements and ensure the app meets user needs.

Long-Term Goals (Post-Hackathon):



Full-Scale Deployment:

- Objective: Expand the deployment of AirWatch.ai to cover multiple cities in Kazakhstan within six months.
- Metrics: Aim for at least 1,000 active users across major cities such as Almaty, Shymkent, and Astana.
- Outcome: Establish AirWatch.ai as a reliable tool for air quality monitoring in Kazakhstan, with widespread adoption among urban residents.

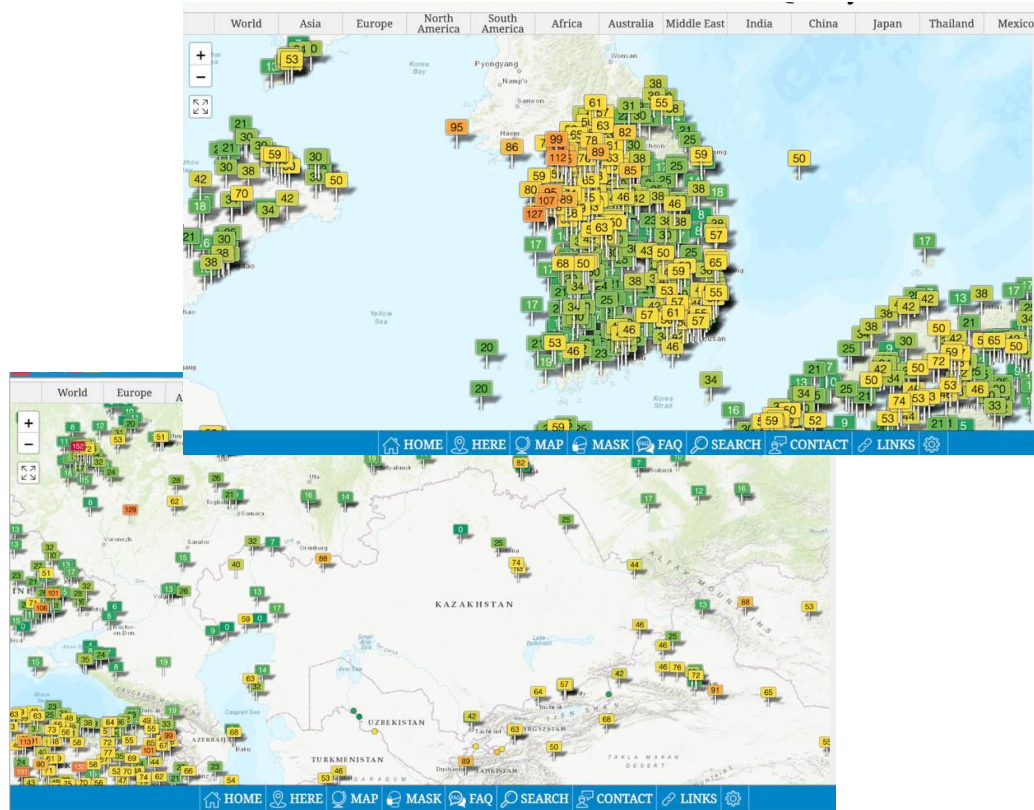
Integration with Municipal Air Quality Programs:

- Objective: Collaborate with local governments to integrate AirWatch.ai with city-wide air quality management strategies.
- Metrics: Secure partnerships with at least two city administrations to use AirWatch.ai data for public health initiatives and pollution control measures.
- Outcome: Contribute to measurable improvements in air quality and public health outcomes by providing actionable data to authorities.



Competitors

1. **PICOCARE:** This app, developed by the Korea Institute of Science and Technology (KIST), uses AI to assess personal exposure to air pollution. It helps users manage their exposure to pollutants by providing real-time air quality data and personalized health tips.
2. **MiseMise:** A mobile app that offers information on air quality and provides tailored health recommendations based on individual health profiles and current pollution levels.
3. **Air Korea:** This is a widely-used app developed by the Korean Ministry of Environment, providing real-time air quality data, including levels of fine dust (PM10, PM2.5), ozone, and other pollutants. It uses AI to predict air quality and provide health recommendations.



Product: demo

Prompt to Gemini AI look like:

You are an AI weather assistant. Based on the following geolocation (latitude: \${lat}, longitude: \${lon}), provide:

Provide the weather forecast in this format:

<h2>Today's weather by current time</h2>
Location: User's city and street

<h3>Current conditions</h3>

- Temperature: (in °C)
- RealFeel temperature: (in °C)
- Wind speed: (in km/h)
- Max Wind Gusts: (in km/h)
- Humidity: (as a percentage)

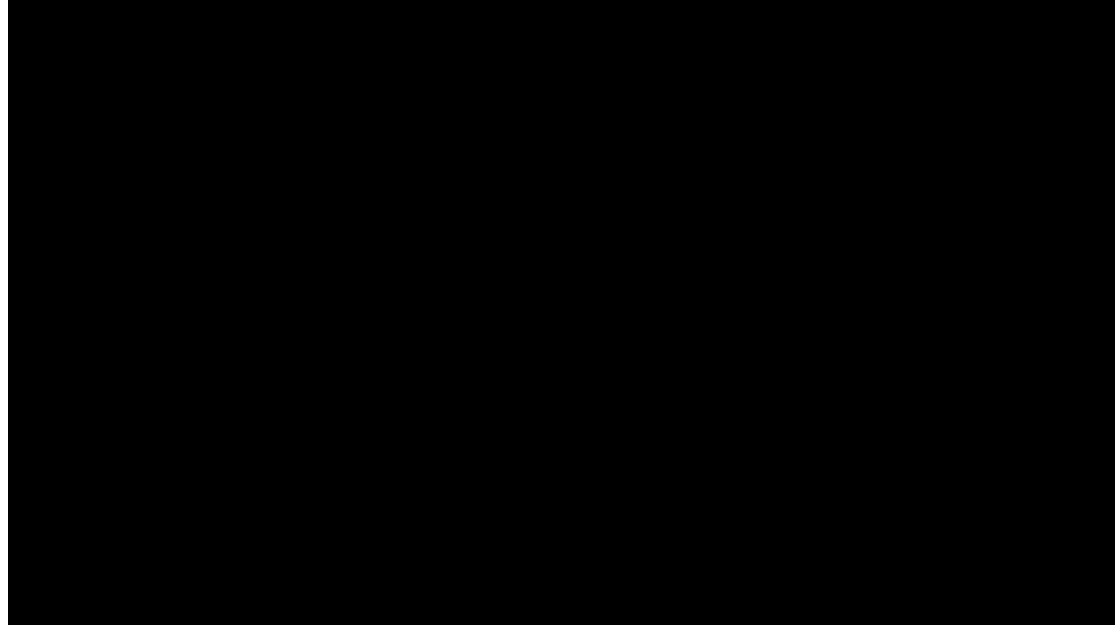
<h3>Probability of Rain, Strong Wind, and Extreme Heat</h3>

- Rain: (in percentage)
- Strong wind: (in percentage)
- Extreme heat: (in percentage)

<h3>Recommendations</h3>

Outdoor Activities: (Advice for outdoor activities)
Health Precautions: (Health-related advice)

Please format your response in HTML, using appropriate tags for headings and paragraphs. Keep the response concise but informative.



Product: short term goals during hackathon & output

Performed short term goals

Name	Result	Comment
Develop a fully functional prototype of the AirWatch.ai web app.	Done	We made the MVP of the product
Complete the integration of Gemini technology for real-time air quality monitoring and predictive analytics.	Done	Yes, connected Gemini AI API key with JS code, and it gives necessary response
Demonstrate the app's capability to provide real-time air quality data and issue alerts to users based on current pollution levels in a selected city (e.g., Almaty).	In process, 80%	At the moment, it works & tested only in Astana. For checking the data from Almaty, we published it on a product developer account for the Chrome Web Store
- Objective: Conduct a small-scale pilot test with a target group (e.g., 100-200 users in a specific area). - Metrics: Track the number of users who sign up and actively use the app during the hackathon. - Outcome: Gather initial user feedback to identify potential improvements and ensure the app meets user needs.	In process, 15%	Collecting only google questionnaire about problem

Desirable output: am I solving the challenge?

- **Improved public health awareness** by delivering timely air quality updates and personalized recommendations for outdoor activities.
- **Reduction of exposure to hazardous weather conditions**, contributing to the sustainability and well-being goals outlined in the project.
- Establish AirWatch.ai as a trusted tool for **real-time environmental monitoring**, driving further adoption in urban areas.



Content

01

Problem

02

Product

03

Architecture

04

Team



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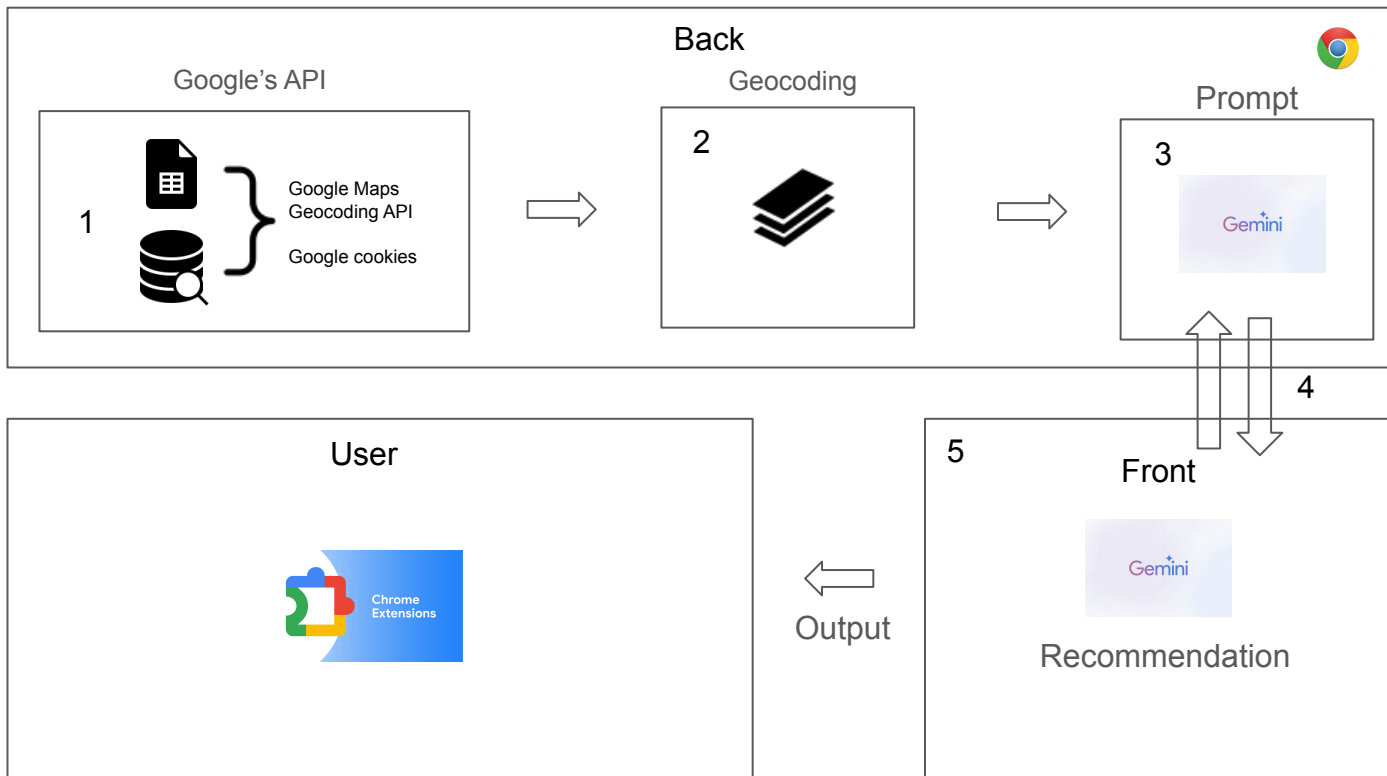
Architecture

Frontend:

1. HTML/CSS/JavaScript: Extension interface that displays weather, air quality, and advisories.
2. Chrome Extension API: Used to integrate with the browser, retrieve geolocation data, and create the extension interface.
3. Popup Window: This window appears when you click on the extension icon and shows information gathered from external APIs and AI.

Backend:

1. Obtaining geolocation: Via browser API (Geolocation API).
2. Requests to OpenWeatherMap and Google Maps Geocoding API: Processed on the backend to provide weather and location data.
3. Processing through Gemini AI: Weather data is passed to Google Cloud AI to analyze and generate recommendations.
4. Update every 60 minutes: Automation process via Chrome Alarms API.
5. Data monitoring and analysis: The collected data is analyzed to improve the service and expand its capabilities.



Code Testing and Challenges

1. The accuracy of real-time weather data

Problem: Gemini AI was providing weather data that sometimes conflicted with real-time data from other sources like OpenWeatherMap. This led to discrepancies in the weather conditions shown to the user.

Resolution: To resolve this issue, we decided to separate the weather data retrieval process from the AI-driven recommendation generation. Instead of relying on Gemini AI for weather data, we integrated the OpenWeatherMap API to fetch real-time weather information (temperature, wind speed, humidity, etc.) and Google Maps Geocoding API for accurate location data.

2. Lack of air quality detectors in Kazakhstan.

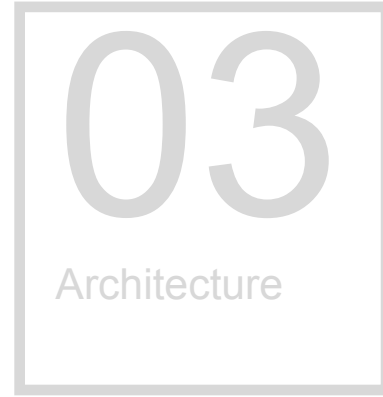
Problem: Many regions, especially in smaller cities, do not have enough detectors to provide real-time, reliable air quality information. This created gaps in the data, potentially affecting the accuracy of recommendations based on air pollution levels.

Resolution: To address this challenge, we implemented a strategy to supplement the limited local air quality data by integrating data from satellite-based APIs and regional weather stations. This approach allowed us to estimate air quality in areas where physical detectors were sparse or unavailable. We used interpolation techniques to estimate air quality based on nearby detector readings and satellite imagery, ensuring that even users in regions with few sensors could receive reasonably accurate air quality information.

GitHub: AirWatch.ai (public)



Content



Team



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Team & accolades



Zhulbarissov Zhassulan
Data Analyst Consultant,
Advisory, Technology department
KPMG Caucasus and Central Asia

Qualification

- Kaspi LAB Certificate in Database Administration
- Kaspi LAB Certificate in Product Analytics
- Datanomix school certificate in Qlik Sense
- Bachelor's Degree in Law, Akhmet Baitursynov Kostanay State University

Experience

- Joined KPMG in 2023. Prior to that, he worked in financial technology companies as a data analyst for over 1 year:
- Participated in project work on implementation of enterprise storage and transactional database.
 - Participated in a project to develop a budgeting and accounting system for a government enterprise.
 - Participated in the project on development of financial analysis and planning for one of the largest GSM companies in Kazakhstan
 - He was involved in IT architecture assessment of a telecom company in Kazakhstan

Achievements

- Finalist in the startup competition "Tayelsizdik Urapgy". Prize fund 100 000 tenge (2021);
- Captain of the KSU Enactus Team 2022;
- Winner of the International scientific - theoretical conference among students and undergraduates "Modern Jurisprudence and Youth" x2 2020, 2021;
- Winner of the annual national competition on data analysis Datathon 2023. Prize fund 500 000 tenge (2023).



Abilgaziev Anuar
Data Science Senior Consultant,
Advisory, Technology department
KPMG Caucasus and Central Asia

Qualification

- Stanford Data Science Leadership Program Certificate. Certificate for successful completion of the online course "Data Scientist with Python Track"
- Republican Mathematics Olympiad, 3rd place
- Siberian Mathematical Olympiad in Novosibirsk, 2nd place.
- Qwant (QWASAR, Silicon Valley), Data Science. Upskill Program training grant.
- Master's degree in Mechanical Engineering, National University of Singapore
- Bachelor's degree in Mechanical Engineering, Nazarbayev University.

Experience

He joined KPMG in 2024. Prior to that, he worked for over 8 years in companies in the fields of Railway Infrastructure, Research and Banking, as an engineer and Head of Data Analysis:

- Working as a systems engineer, improved mechanical properties of printed objects by 40% and accuracy of printed objects by 10%. Also, reduced surface roughness by 20%.
- Working as a mechanical engineer at the French Engineering Company, he participated in the project for the design of a new electric locomotive repair shop.
- He was engaged in scientific activity in the School of Engineering and Digital Sciences, Nazarbayev University, projects on "multi-material and multi-color 3D printer FDM-type". Wrote and published 9 research articles and a book section in scientific papers.
- Managed the data analysis department in one of the leading banks in Kazakhstan. The department created machine learning models for automatic calculation of provisions (reserves), calculation of amounts for state premium payments on deposits in tenge, anti-fraud system for calculating the degree of probability of fraud among managers in branches.



Contacts

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LinkedIn: Anuar Abilgaziev



Desirable results in hackathon



Customers



Revenue model



Roadmap & GTM

Workshop Details

Workshop 1: August 24, 2024

Zoom Link: Join Workshop 1 - <https://strategieast.zoom.us/j/83316187084>

Almaty Time: 16:00

Topic: Technical Assessment Criteria

Led by: Askar Aituov, Google Developer Relations Coordinator

Description: Gain insights into the technical evaluation criteria used to assess project submission.

Workshop 2: August 30, 2024

Zoom Link: Join Workshop 2 - <https://strategieast.zoom.us/j/82242921268>

Astana Time: 16:00

Topic: How to Deploy Applications on Google Cloud

Led by: Enes Turan, Google Cloud Expert

Description: Learn how to effectively deploy your AI-powered solutions on Google Cloud.

Workshop 3: September 7, 2024 - <https://strategieast.zoom.us/j/84381608743>

5. Is there evidence of next steps? Does the team display a clear plan for future extension to a larger audience if they were to continue? (5 POINTS)



Market size



financial & projections



Consider

3. Feedback from users, testing & iteration (5 POINTS)

- a. Does the team clearly describe three feedback points they received from real users and the steps they took to test them?
- b. Is there evidence of what the team learned and how the solution was iterated upon based on user feedback?

A good problem statement:

- Presents an opportunity of some significance, urgency, and priority
- Relates in the simplest terms possible, the process of concern and impact on the community
- Includes enough data so the audience can understand the size and scope of the problem, together with the degree or magnitude of the problem
- Should not be more than one paragraph (as little as 2-3 sentences, as much as 5-6 sentences)

How to structure a problem statement

- Who is experiencing the problem?
- What segment/s of the community, how many?
- What is the problem?
- Describe issue being experienced or need going

3. Code testing and iteration (5 POINTS)

- a. Does the team highlight one challenge they faced while building their code, how they addressed the issue, and the technical decisions and implementations they had to make? Did the team include guidance on running their code?

4. Does the video demonstration show an actual working application (not a mockup) and how a user will interact with the solution? Does the demo show how the solution makes effective and appropriate use of the features of the chosen Google technology or platform? (5 POINTS)
5. Can the solution, in its current form or minor structural changes, support more users and increased scale? (5 POINTS)

Criteria for admission: Презентация для пичта изменится * (если пройду)

Impact (25 Points)

1. **Challenge Definition** (5 points): Clearly define the problem with a concise problem statement.
2. **Alignment with UN SDGs** (5 points): Identify the relevant SDG(s) and explain how your solution addresses them.
3. **User Feedback and Iteration** (5 points): Provide feedback from real users, describe the testing process, and demonstrate iteration based on this feedback.
4. **Problem-Solution Fit** (5 points): Use metrics, goals, and outcomes to show how your solution addresses the defined challenge.
5. **Scalability and Future Steps** (5 points): Present a clear plan for scaling the solution to a wider audience.

1. **4,5,7 slide.** The challenge is to provide real-time, location-specific air quality and weather insights along with health and activity recommendations, using a Chrome extension that using Gemini AI.
2. **7 slide.** I have chosen:
 - a. SDG 3: Good Health and Well-being
 - b. SDG 11: Sustainable Cities and Communities
 - c. SDG 13: Climate Action
3. **10 slide.** In process
4. **10 slide.** In process
5. **7 slide.** I decided to make a long tasks
 - a. Full-Scale Deployment: Aim for at least 1,000 active users across major cities such as Almaty, Shymkent, and Astana.
 - b. Integration with Municipal Air Quality Programs: Collaborate with local governments to integrate AirWatch.ai with city-wide air quality management strategies.

Technology (25 Points)

- **Technical Architecture** (5 points): Describe the architecture, high-level components, and use of Google technology relevant to your solution.
- **Implementation Completeness** (5 points): Ensure all technical components necessary for solving the challenge are implemented.
- **Code Testing and Challenges** (5 points): Highlight a coding challenge you faced, how it was resolved, and any key technical decisions.
- **Demo Presentation** (5 points): Include a video demo showcasing user interaction and effective use of Google technology.
- **Scalability** (5 points): Demonstrate how your solution can scale to support more users effectively.

1. **12 slide.**
2. **X slide.** I don't know
3. **13 slide.**
4. **9 slide.** Процесс идет на стороне одобрения аккаунта разработчика продуктов для Интернет-магазина Chrome. Ожидаем как пройдет и сможем запустить его в прод. Думаю как можно показать это в виде видео.
5. **9 slide.** Это гугл расширение использующее токены Gemini AI. Масштабирования среди юзеров идет бесшовно, хватит лишь использовать веб браузер. На приложении в телефоне пока под вопросом.

